

Sales Data Analysis and Business Intelligence Report

Comprehensive Data Cleaning, Integration, and Insight Extraction

1. Introduction

In modern business environments, data-driven decision-making has become essential for strategic growth and operational efficiency. This project aims to analyze a multi-table retail sales dataset to extract meaningful business insights related to customers, orders, products, and geographical locations.

The analysis focuses on understanding profitability patterns, shipping performance, regional trends, and the impact of discounts on financial outcomes.

2. Project Objective

The main objectives of this project are:

- Clean and standardize raw retail datasets.
- Integrate multiple data sources into a unified analytical dataset.
- Detect anomalies and outliers affecting business metrics.
- Discover sales and profitability patterns.
- Identify relationships between operational variables.
- Generate actionable business insights.

3. Dataset Description

The project integrates four datasets:

1. Customers.csv
2. Location.csv
3. Orders.csv
4. Products.csv

Each dataset represents a different business entity:

- Customers → Client information
- Location → Geographic distribution
- Orders → Sales transactions
- Products → Product classification

4. Data Cleaning and Preprocessing

Data cleaning was a crucial stage to ensure analytical accuracy and reliability.

4.1 Customers Dataset Cleaning

Steps Performed:

- Checked for null values.
- Verified uniqueness of Customer ID.
- Checked duplicated customer names.
- Standardized column names (lowercase, underscores).
- Cleaned customer names (strip + title case).

Why This Matters:

- Ensures referential integrity before merging.
- Prevents duplication during joins.
- Improves data consistency for reporting.

4.2 Location Dataset Cleaning

Steps:

- Standardized column names.
- Checked missing values.
- Verified duplicates.
- Analyzed postal code uniqueness.
- Visualized distribution by region.
- Identified top 10 cities by record count.

Insights Gained:

- Some regions dominate the dataset.
- Certain cities represent major customer clusters.

Business Value:

Helps identify geographic sales concentration and potential market expansion opportunities.

4.3 Orders Dataset Cleaning

Steps:

- Standardized column names.
- Removed duplicate records.

- Converted order_date and ship_date to datetime format.
- Created new feature: ship_days.
- Detected outliers in sales using IQR.
- Visualized sales and profit distribution.
- Generated histograms and boxplots.

4.3.1 Outlier Detection (Sales)

Using IQR:

Outliers were detected beyond $1.5 \times \text{IQR}$ range.

Why Important?

- Extreme sales values distort averages.
- Outliers may represent data entry errors or exceptional bulk orders.

4.3.2 Shipping Duration Feature Engineering

$\text{ship_days} = \text{ship_date} - \text{order_date}$

This allowed:

- Measuring operational efficiency.
- Studying delivery performance.
- Linking shipping time to profitability.

4.3.3 Sales Over Time Analysis

Sales aggregated yearly revealed:

- Growth trends over time.
- Business expansion phases.

- Revenue consistency patterns.

4.3.4 Shipping Duration vs Sales

Scatter plot analysis revealed:

- Weak correlation between longer shipping and higher sales.
- Shipping delays do not significantly increase revenue.

Business Insight:

Operational delays do not justify higher order value.

4.4 Products Dataset Cleaning

Steps:

- Standardized column names.
- Removed unnecessary unnamed columns.
- Renamed sub-category to sub_category.
- Removed null rows.
- Verified duplicates.

Why This Is Critical:

Product classification drives profitability analysis and regional category performance studies.

5. Data Integration

All datasets were merged:

orders → customers → location → product

Result: final_df

This unified dataset enabled multidimensional analysis combining:

- Financial metrics
- Geographic data
- Customer segmentation
- Product categories

6. Exploratory Data Analysis and Insights

6.1 Profit by Region

Bar plot showed clear regional profit variation.

Key Insight:

Some regions generate significantly higher profits, suggesting stronger market penetration or better pricing strategies.

Business Use:

- Focus investment on high-profit regions.
- Reevaluate performance in low-profit areas.

6.2 Most Sold Categories

Countplot revealed category dominance.

Insight:

Certain product categories drive the majority of sales volume.

Business Value:

- Inventory planning
- Marketing prioritization
- Supply chain optimization

6.3 Shipping Mode Efficiency

Average ship_days per shipping mode showed:

- Faster modes reduce delivery time significantly.
- Standard shipping remains dominant.

Business Implication:

Premium shipping may not always justify cost unless tied to high-margin products.

6.4 Impact of Discounts on Profit

Line plot between discount and profit revealed:

- Higher discounts often push profit toward zero or negative.
- Aggressive discounting reduces financial sustainability.

This is one of the most critical discoveries.

Business Insight:

Discount strategies must be optimized. Over-discounting damages profitability.

6.5 Category Distribution by Region

Countplot (region $\times$ category) revealed:

- Regional purchasing preferences differ.
- Some regions specialize in specific categories.

Business Use:

Localized marketing strategies.

7. Profit Outlier Removal

Profit range restricted between:

-2000 and 5000

Why?

Extreme losses or extreme gains distort overall profitability averages.

After filtering:

Analysis became more realistic and representative of typical business behavior.

8. Correlation Analysis

Numerical variables analyzed:

- Sales
- Quantity
- Discount
- Profit
- Ship_days

Heatmap Findings:

- Strong positive correlation between sales and profit.
- Negative correlation between discount and profit.
- Weak correlation between ship_days and profit.

Key Insight:

Discount is the most sensitive profitability driver.

9. Regional and Category Profit Heatmap

Pivot table analysis showed:

- Some regions perform better in specific categories.
- Not all categories are equally profitable across regions.

Strategic Value:

Supports region-specific product portfolio optimization.

10. Patterns Discovered

Through analysis, the following patterns emerged:

1. Profit is strongly dependent on discount levels.
2. Shipping speed does not strongly influence revenue.
3. Certain regions dominate overall profitability.
4. Product category demand varies by region.
5. Sales growth shows temporal expansion trend.
6. Outliers significantly distort financial indicators.
7. Operational efficiency measured via ship_days reveals logistical performance levels.

11. Real-World Applications

This analysis can be used for:

- Pricing strategy optimization.
- Discount policy refinement.
- Regional market expansion planning.

- Logistics performance improvement.
- Inventory management.
- Category-based marketing strategies.
- Risk reduction from unprofitable discounting.

12. Business Impact

From Data Cleaning:

Ensured reliability and eliminated misleading metrics.

From Feature Engineering:

Enabled operational efficiency analysis.

From Integration:

Created 360° business view.

From Visualization:

Made decision-making intuitive and evidence-based.

From Correlation Analysis:

Identified key profit drivers.

13. Overall Contribution

This project demonstrates full-cycle business analytics:

Raw Data → Cleaning → Transformation → Integration →
Visualization → Insight → Strategic Interpretation.

It converts raw transactional data into strategic business intelligence.

14. Professional Conclusion

This project successfully transformed fragmented retail datasets into an integrated analytical framework capable of delivering actionable business insights.

The analysis revealed that:

- Discount management is the most critical profitability factor.
- Geographic performance varies significantly.
- Operational efficiency impacts customer experience but not necessarily revenue.
- Category distribution differs by region, requiring localized strategies.

By applying structured data cleaning, rigorous outlier detection, feature engineering, and correlation analysis, we ensured analytical integrity and extracted high-value insights.

The ultimate benefit of this analysis lies in its ability to support strategic decision-making, improve financial performance, and optimize operational efficiency.

This work demonstrates how systematic data analysis directly translates into measurable business intelligence and competitive advantage.

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